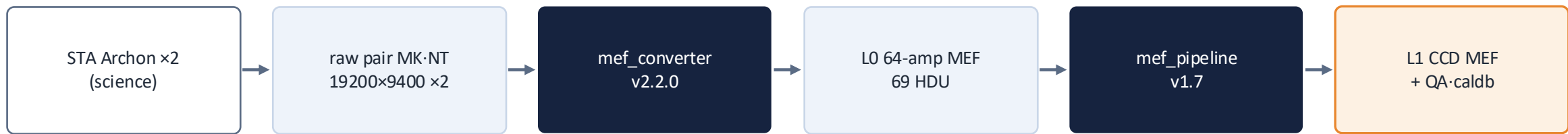
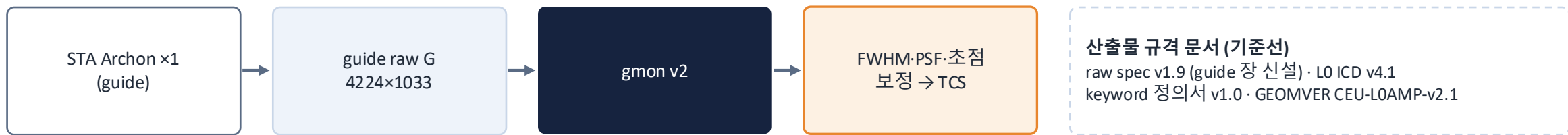


전체 그림 — Archon raw부터 L1 · 시상 모니터까지 한 체계로

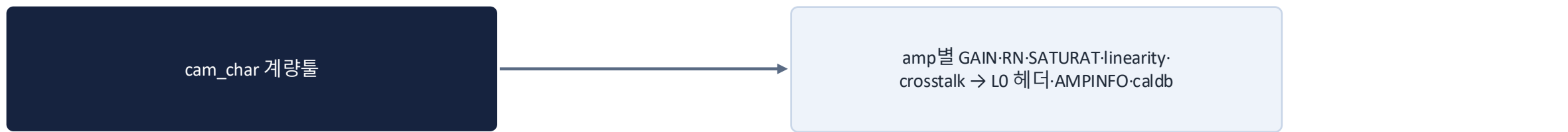
과학 노출 (노출 1회 = raw 2개)



가이드 노출 (시상 모니터)



특성 측정 (실험실·사이트 캠페인)



■ 신규 구현 코드 ■ 데이터 제품 ■ 최종 산출물 □ 하드웨어

→ 실측되면 코드 수정 없이 값만 교체 (D-005)

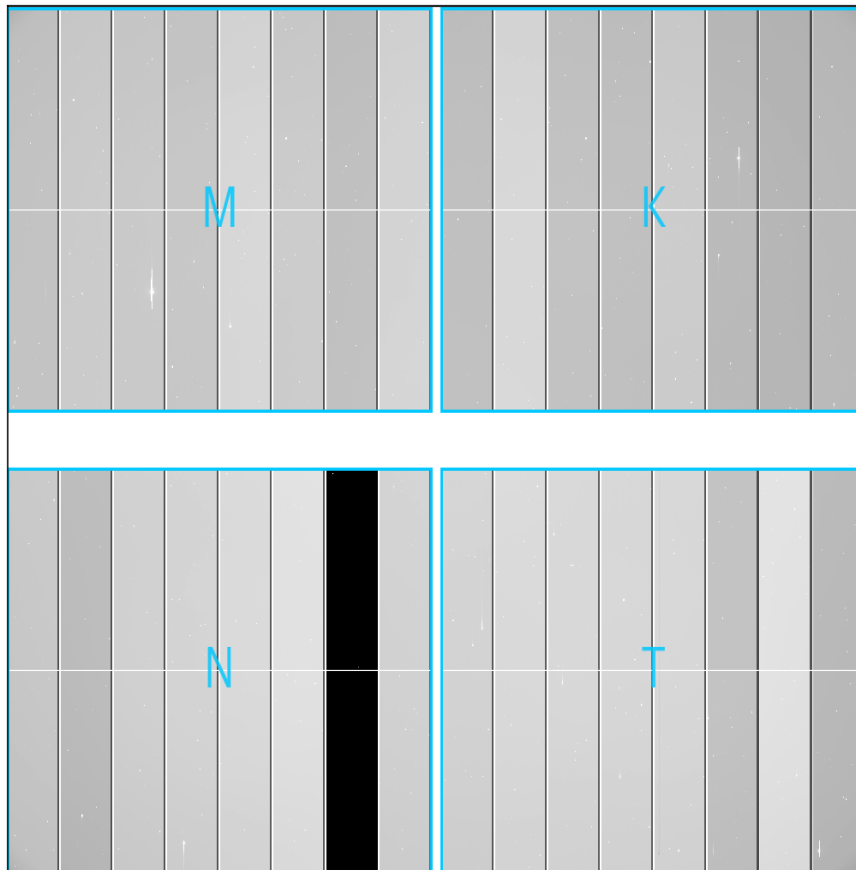
- 1 MEF·raw 규격 확정
raw v1.9 · ICD v4.1 · 변환기 v2.2.0
- 2 전처리 v1.7 검증
mock 야간 40노출 end-to-end
- 3 기준선 리포트 v1.0
3사이트 449 프레임 실전 측정
- 4 gmon v2 재구현
합성·실프레임 주입 시험 통과

1 64-amp MEF 포맷 정의 — 입력(raw)·출력(L0) mef_converter

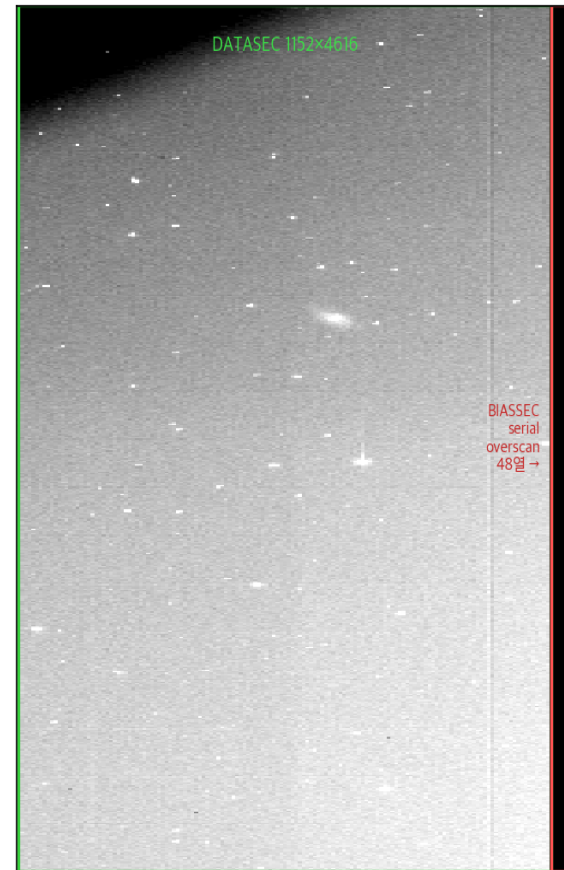
- MEF 규격 확정: **L0 MEF ICD v4.1** + keyword 정의서 v1.0 — md·docx 병행 관리
- 규격에 맞춰 **Archon raw spec v1.9** (MK·NT pair + guide G 장 신설)
- L0 영상 구성: **69 HDU** = amp IMAGE 64장 (1200×4616, 활성 1152 + overscan 48 열) + BINTABLE 4 (AMPINFO·XTALKINFO·VOLTINFO·TELEMETRY)
- Geometry: **AMPINFO 테이블** (GEOMVER CEU-LOAMP-v2.1) — 파이프라인·계량틀이 하드코딩 없이 같은 표를 읽음
- 변환기 v2.2.0: raw pair → L0 (사이트 코드 파일명, D-011) · 구형 32-amp → mock64 변환기 병행
- 검증: DATASEC trim 조립 mosaic이 구형 원본과 **4개 CCD 픽셀 단위 완전 일치** (K/N 역순 strip 매핑 포함)

HDU 69개 = PRIMARY + IMAGE×64 (amp raw) + BINTABLE×4 (AMPINFO · XTALKINFO · VOLTINFO · TELEMETRY) | GEOMVER CEU-LOAMP-v2.1

L0 64-amp MEF 한 노출 — 초점면 4 CCD × 16 amp (저장 상태 그대로, OBJECT I 60 s)



amp HDU 1개 (M01T, 1200×4616 uint16)

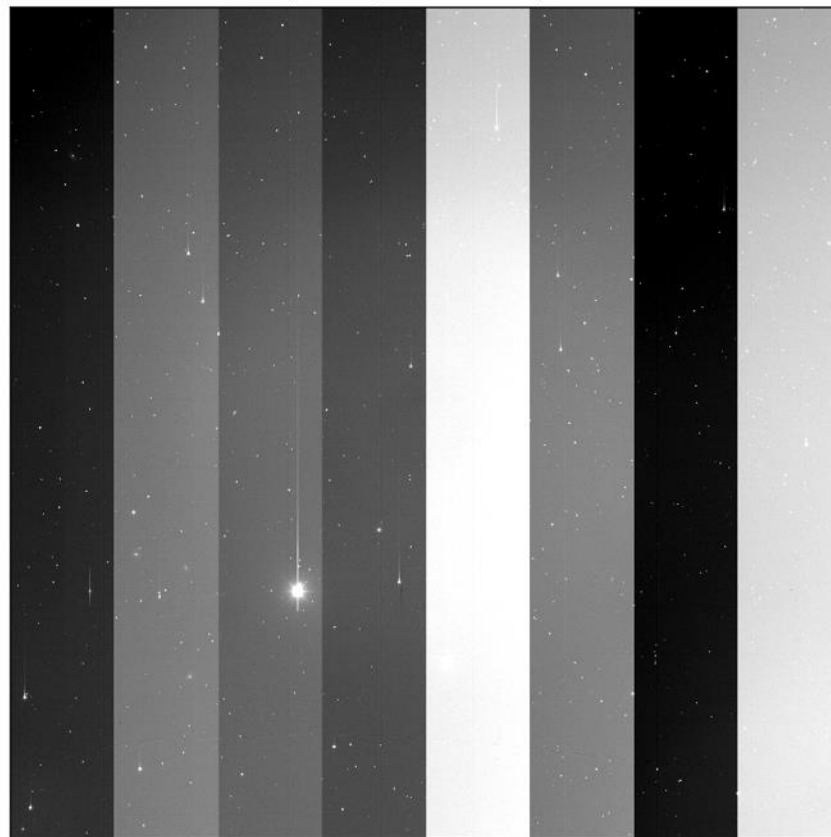


mock 노출의 L0 실파일 렌더 — 좌: 초점면 4 CCD × 16 amp 타일(죽은 N칩 채널이 검게 그대로 보임), 우: amp HDU 구조(DATASEC·serial overscan)

L0 → L1 전처리 파이프라인 v1.7 — mock 으로 전 구간 검증

- **19단계** (overscan → bias → gain → flat → fringe → illum → BPM → AMPMATCH → 조립 → CR → sky → astrometry → ZP) — 순수 Python·NumPy·astropy, 단위테스트 64
- mock 관측영상 end-to-end: OBJECT **28장 전량 L1 생성**, 구조·provenance 검사 통과
- amp 경계 단차 **>100배 감소**: $470\text{--}6,178\text{ e}^- \rightarrow 2.2\text{--}4.2\text{ e}^-$ (28노출 중앙값 $23\text{ e}^- \approx \text{sky의 } 0.5\%$)
- Astrometry: Gaia DR3 **TAN-SIP(3)** 105칩 해, rms 중앙값 $0.31''$ · 로컬 Gaia 스토어로 오프라인 성표 $<0.5\text{초/노출}$
- 측광 ZP: GSPC 합성 JKC V/I 기준 **ZPRMS 0.035–0.041 mag** — 투명도 변화를 수십 mmag로 추적
- QA가 실제 경험값을 자동 검출(죽은 N칩 채널, overscan 하늘 오염) · **~장당 24초** 소모 (4병렬, 서버 배포 가이드 포함)

CCD M, kmtc.20260630.011103 (mock 야간 OBJECT I 60 s) — amp seam >100배 감소 ($470\text{--}6,178\text{ e}^- \rightarrow 2.2\text{--}4.2\text{ e}^-$)
 교정 전: L0 amp 조립만 (ADU)
 — amp별 bias·gain 차이로 16-amp 줄무늬



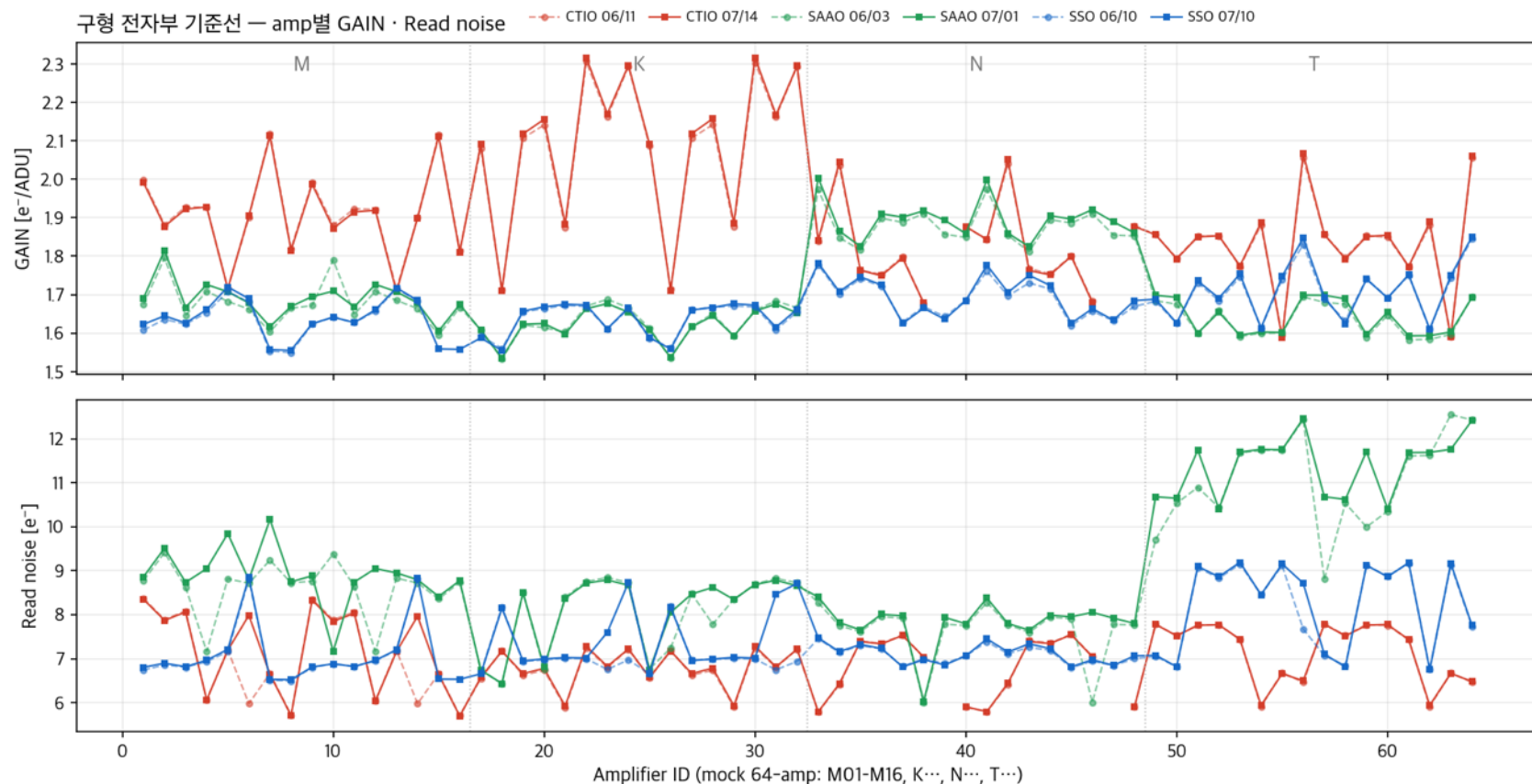
L1 (v1.7 전처리 후, electrons)
 — overscan·bias·flat·fringe·illum·AMPMATCH



같은 노출(CCD M, I 60 s)의 교정 전 amp 조립(좌)과 L1(우) — mock은 실제 legacy 픽셀 기반이라 fringe·오염 등 실제 현상이 재현

3 카메라 특성 계량툴 + 구형 전자부 기준선 리포트 v1.0

- 계량툴 **kmt_cam_char**: QC census · RN · PTC(gain) · linearity · EPER(CTE) · PRNU · 리포트 자동 생성 — 파이프라인의 L0 리더 · geometry를 재사용(교차검증)
- Archon 실험실 취득 스크립트 **v2.0** (캠페인 INI·시뮬레이터 테스트) — 신규 전자부 측정 준비 완료
- 첫 캠페인 LEGACY-2026-06/07: **3사이트 × 2 야간 · 449 프레임**으로 측정 코드 실전 검증 + 신규 비교 기준선 확보
- 기준선(중앙값): **GAIN 1.67–1.88 e⁻/ADU · RN 7.0–8.7 e⁻** · PRNU ~0.9% · TOP/BOT 일치 0.1–0.2% · 두 달 gain 재현성 ~0.2%
- 신규 전자부 **pass/fail 기준(ACT-005)의 입력** — 측정값은 코드 수정 없이 L0 헤더 · AMPINFO·caldb로 주입

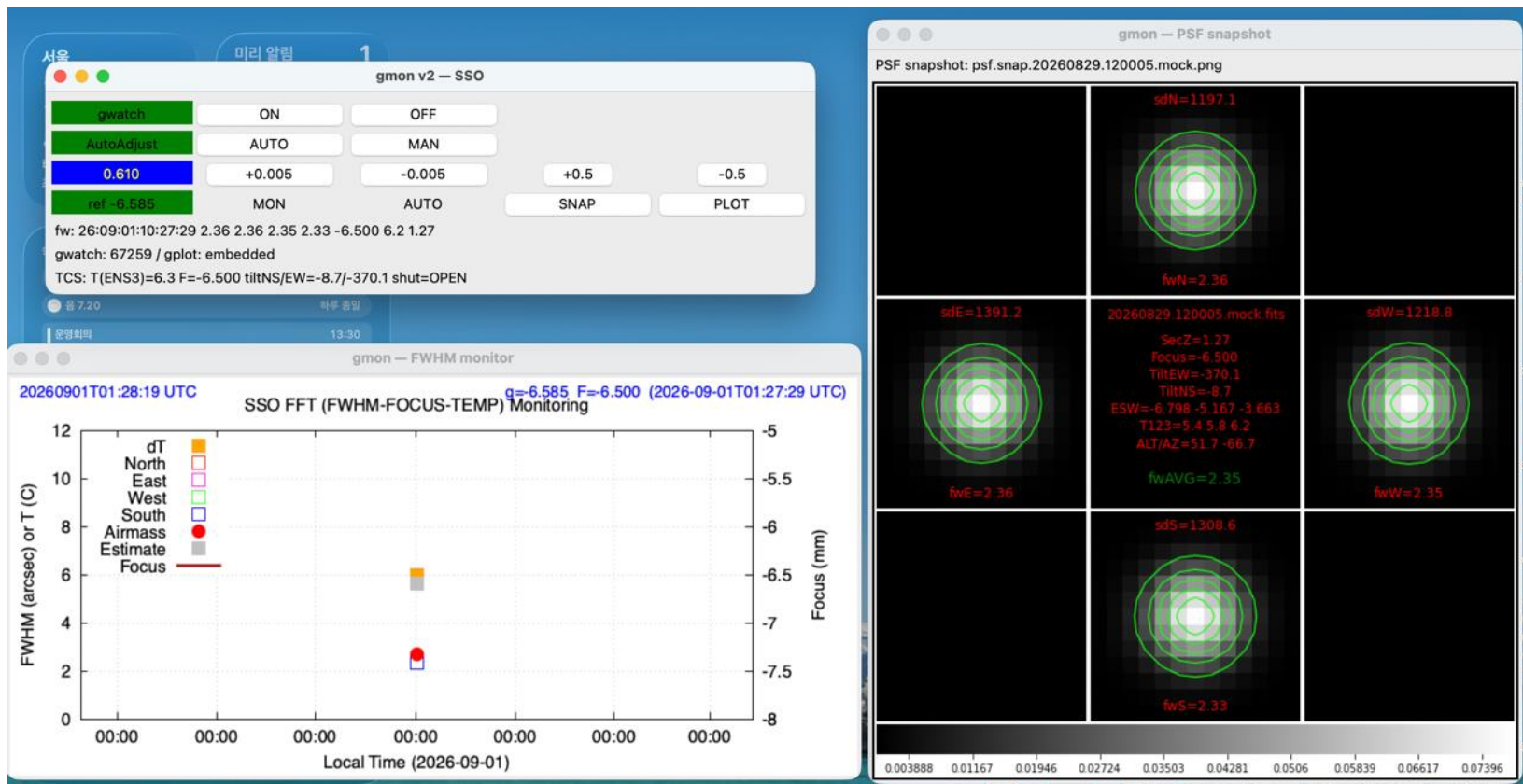
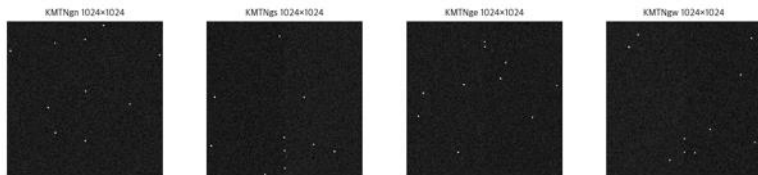


6개 캠페인(3사이트 × 2야간)의 amp별 GAIN·RN — 사이트 별 특성(CTIO gain 패턴, SAAO T칩 RN 상승)과 재현성. 끊긴 점은 죽은 amp

Archon 가이드 카메라 기반 gmon 재구현 (v2)

- 레거시(2018, wish·Perl·IRAF) → **Python 전면 재작성**:
gwatch→gsplit→gpsf(sex·psfex)→gsnap→gplot + Tkinter GUI 3창
- 신규 **gsplit**: Archon 단일 FITS(4224×1033, CCD47-20 4 칩×2채널) → 칩별 4장 — 채널 스티칭·페데스탈 단차 보정
- 온도 기반 초점 보정 **ref = slope·T - (base+dfocus)** → TCS 전송, AUTO/MAN·안전 범위·max_jump·dry-run 안 전장치
- 레거시 화면·파일 형태 유지 + 오류 수정(fwAVG 평균 버그, 원본 삭제 처리 필요: 파일사이즈 압박)
- 합성·실프레임 **별 주입 시험 통과** · TCS 시뮬레이터로 초점 루프까지 로컬 검증
- PSF 찌그러짐 방향성 → 90도씩 회전(현장 최종 확인)

gsplit: 단일 FITS → 칩별 4장 (채널 스티칭 + 페데스탈 단차 보정) → sex/psfex FWHM → 스냅샷·그래프
가이드 raw 1장 (Archon, 4224×1033 unit16) → CCD47-20 4칩 × 2채널 — 합성 별 40개 주입 시험



현장(커미셔닝)에서 확정할 항목 — DESIGN §10

칩↔방위 매핑·표시 회전각 / 픽셀 스케일(0.52" 가정) / 더미 9행·프리스캔 위치 / TCS fttgoto 검증 후 dry_run 해제

SAOImage ds9

File Edit View Frame Bin Zoom Scale Color Region WCS Illustrate Analysis Help

File kmtk.20260827.285410.ceu.i0amp.v2_1_1.mef.fits[K06T]

Object DS2004

Value

WCS

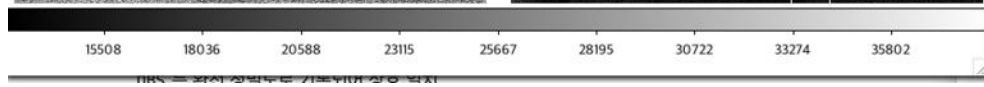
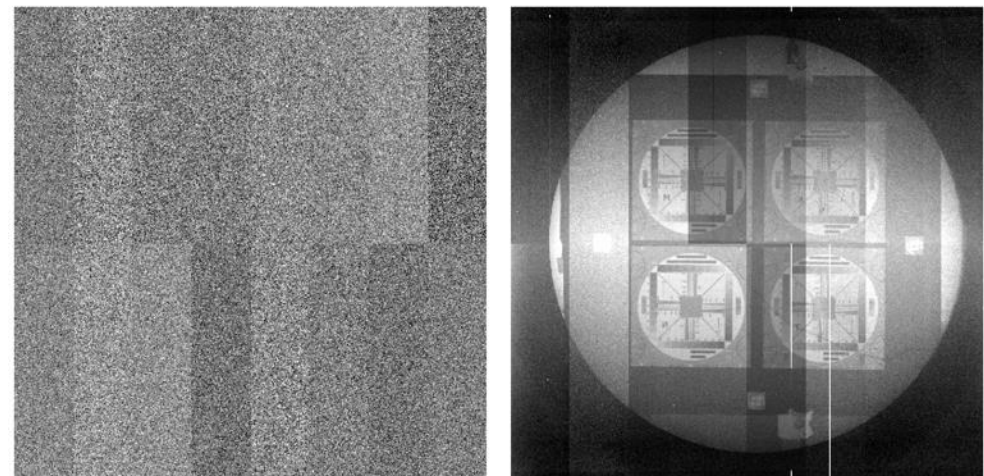
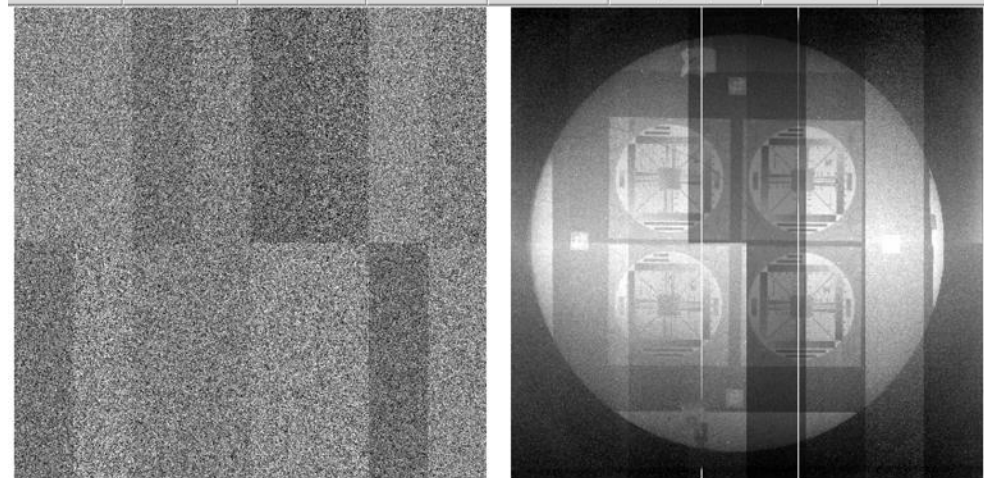
Physical x y

Image x y

Frame 1 x 0.0439759 0

file edit view frame bin zoom scale color region wcs illustrate analysis help

open save header notes samp page setup print exit



kmtk.20260827.285410.ceu.i0amp.v2_1_1.mef.fits

File Edit Font

```
TIMCONF = 'CEU TIM v1.0' / CCD clock and timing configuration
CTRLVER = 'ARCHON-v1.0' / controller system version
TIMVER = 'TIM-v1.0' / timing script version
XTALKVER = 'UNMEASURED' / crosstalk model version
XTALKCAL = 'F' / crosstalk coefficients are placeholders
BIASVER = 'BIAS-v1.0' / bias configuration version
CLKVER = 'CLK-v1.0' / clock configuration version
PIPEVER = 'kmt_ceu_i0amp_mknt2mef-v2.2.0' / converter version
REFVER = 'N/A' / reference image version
CATVER = 'N/A' / catalog version
COMMENT TCS pointing information
TCSLINK = 'NC' / TCS communications link status
TCSARC = 'NC' / TCS auto recovery mode status
TCSQDATE = 'NC' / UTC date/time of last TCS query
TCSUDATE = 'NC' / UTC date/time of last TCS update
TIMESYS = 'UTC' / time system
DATE-OBS = '2026-08-27T02:40:55.417' / UTC date/time at start of observation
MJD-OBS = 61279.11175251147 / modified Julian date at start
JD = 2461279.6117525115 / Julian date at start
UT = '2026-08-27T02:40:55.417' / UTC timestamp
RADECSYS = 'NC' / telescope coordinate system
RA = 'NC' / telescope RA
DEC = 'NC' / telescope DEC
EQUINOX = 'NC' / coordinate system equinox
HA = 'NC' / hour angle at start
ST = 'NC' / local sidereal time at start
SECZ = 'NC' / secant of zenith distance
ALT = 'NC' / telescope altitude [deg]
AZ = 'NC' / telescope azimuth [deg]
TCSDRIV = 'NC' / telescope drive status
TELMOVE = 'NC' / telescope motion status
COMMENT Filter/shutter, FSA, focus, dome, and thermal info
AUXLINK = 'NC' / AUX control system communication status
AUXARC = 'NC' / AUX link auto recovery status
AUXQDATE = 'NC' / UTC date/time of last AUX query
AUXUDATE = 'NC' / UTC date/time of last AUX update
FSSTAT = 'NC' / filter-shutter subsystem status
FILTOP = 'NC' / filter operational status
FILNUM = 'NC' / filter selector position number
FILTER = 'NC' / filter name in beam
SHUTOP = 'NC' / shutter operational status
SHUTER = 'NC' / shutter position
FSATEMP = '-999.99' / FSA internal temperature [C]
FSAHUM = '-999.99' / FSA internal relative humidity [%]
FSADEW = '' / FSA internal dew point [C]
FSAALRM = '' / FSA environmental alarm status
FASTAT = 'NC' / focus actuator subsystem status
FAFOCUS = 'NC' / focus position offset [mm]
FATILTNS = 'NC' / focus tilt NS offset [arcsec]
FATILTEW = 'NC' / focus tilt EW offset [arcsec]
FAPOSS = 'NC' / south focus actuator position [mm]
FALIMS = 'NC' / south focus actuator limit status
FAPOSE = 'NC' / east focus actuator position [mm]
FALIME = 'NC' / east focus actuator limit status
FAPOSW = 'NC' / west focus actuator position [mm]
FALIMW = 'NC' / west focus actuator limit status
DSSTAT = 'NC' / dome shutter status
DSUP = 'NC' / upper dome shutter position
DSLW = 'NC' / lower dome shutter position
DSSAF = 'NC' / dome safety status
DSAUTO = 'NC' / dome auto sync status
DSALT = 'NC' / dome slit altitude [deg]
DSAZ = 'NC' / dome slit azimuth [deg]
DSTELALT = 'NC' / telescope altitude used by dome [deg]
DSTELAZ = 'NC' / telescope azimuth used by dome [deg]
DALTERR = 'NC' / dome-telescope altitude difference [deg]
DAZERR = 'NC' / dome-telescope azimuth difference [deg]
MCSTAT = 'NC' / mirror cover status
MCPOS = 'NC' / mirror cover position [%]
CHSTAT = '' / chiller status
ENSTAT = 'NC' / environmental control system status
ENFAN = 'NC' / environmental system fan state
CCDTEMP = '-999.99' / CCD temperature [C]
DEWPRES = '9.99e-9' / dewar pressure
PT30N1 = '-999.99' / cooler temperature sensor 1 [C]
PT30N2 = '-999.99' / cooler temperature sensor 2 [C]
CHARCOAL = '-999.99' / charcoal getter temperature [C]
AIR_IN = '' / air inlet temperature [C]
AIR_OUT = '' / air outlet temperature [C]
GLYC_IN = '' / glycol inlet temperature [C]
GLYC_OUT = '' / glycol outlet temperature [C]
CHKING = '' / image check status
CHKING_C = '' / image check comment
END
```

kmtk.20260827.285410.ceu.i0amp.v2_1_1.mef.fits[M01T]

File Edit Font

```
XTENSION = 'IMAGE' / image extension
BITPIX = 16 / array data type
NAXIS = 2 / number of array dimensions
NAXIS1 = 1200 / amp image width including overscan
NAXIS2 = 4616 / amp image active rows
PCOUNT = 0
GCOUNT = 1
BZERO = 32768 / unsigned 16-bit zero point
BSCALE = 1 / default scale
BUNIT = 'ADU' / pixel unit
EXTNAME = 'M01T' / amplifier image extension
EXTTYPE = 'AMP_RAW' / L0 amplifier raw image
REALDATA = 'T' / actual amplifier data from Archon raw
DATAPROD = 'L0_AMP' / data product type
CHIPID = 'M' / CCD identifier
CCDNAME = 'KMTNet CCD M' / CCD name
AMPID = 1 / global amplifier ID
AMPSEQ = 1 / amplifier sequence within CCD
STRIPID = 1 / vertical strip ID
ENDID = 'TOP' / readout end ID
AMPNAME = 'M01T' / amplifier name
RAWFILE = 'KMTK.20260827.285410.MK.fits' / source raw FITS file
CTRLID = 1 / science controller ID
MODULE = 1 / controller module placeholder
CHANNEL = 1 / controller channel placeholder
CHIPFLP = 'None' / no chip-dependent OSU-style flip applied
STRIPDIR = '+X' / strip number direction in CEU L0 packing
READDIR = '-Y' / physical readout direction placeholder
CCDSUM = '1' / on-chip binning factors
CCDSEC = '[1:1152,4617:9232]' / amplifier section in CCD coords
AMPSEC = '[1:1152,4617:9232]' / amplifier section in CCD coords
DETSEC = '[1:1152,14782:19397]' / amplifier coords on detector mosaic
RAWDATA = '[1:1152,4785:9400]' / source raw data section
RAWBIAS = '[1153:1200,4785:9400]' / source raw overscan section
DATASEC = '[1:1152,1:4616]' / active data section
PRESEC = '[1:0,1:4616]' / no prescan in Archon raw
BIASSEC = '[1153:1200,1:4616]' / local overscan section
TRIMSEC = '[1:1152,1:4616]' / trimmed data section
RAWNAX1 = 19200 / source raw image width
RAWNAX2 = 9400 / source raw image height
RAWXTILE = 1200 / raw amp tile width
AMPDATA = 1152 / active columns per amp tile
OVERSCANX = 48 / X overscan columns per amp tile
PRESCANX = 0 / X prescan columns per amp tile
MIDOVSCY = 168 / middle Y overscan rows ignored
GAIN = 0.0 / gain placeholder [e-/ADU]
RONNOISE = 0.0 / read noise placeholder [e-]
SATURAT = 62000 / saturation level placeholder [ADU]
LINMAX = 58000 / linearity maximum placeholder [ADU]
FILTER = 'NC' / filter name in beam
PROJID = 'ENG' / project ID
IMAGETYPE = 'FLAT' / type of observation
OBJECT = 'DS2004' / object name
OBSTYPE = 'FLAT' / type of observation
RA = 'NC' / telescope RA
DEC = 'NC' / telescope DEC
HA = 'NC' / hour angle
ST = 'NC' / local sidereal time
SECZ = 'NC' / airmass
ALT = 'NC' / telescope altitude [deg]
AZ = 'NC' / telescope azimuth [deg]
UT = '2026-08-27T02:40:55.417' / UTC timestamp
CTYPE1 = 'RA---TAN' / coordinate type
CTYPE2 = 'DEC--TAN' / coordinate type
CRVAL1 = 0.0 / placeholder coordinate reference RA [deg]
CRVAL2 = 0.0 / placeholder coordinate reference DEC [deg]
CRPIX1 = 0.0 / placeholder coordinate reference pixel
CRPIX2 = 0.0 / placeholder coordinate reference pixel
CD1_1 = -0.00010972222222222222 / coordinate transform matrix
CD1_2 = 0.0 / coordinate transform matrix
CD2_1 = 0.0 / coordinate transform matrix
CD2_2 = 0.00010972222222222222 / coordinate transform matrix
WCSDIM = 2 / coordinate system dimensionality
END
```